JUMUIA X

DEVELOPMENT OF AN INTEGRATED SERVICE BOOKING
AND BUSINESS MANAGEMENT INFORMATION SYSTEM

BASELINE PROJECT PLAN (BPP)

Course	INSY281 — System Analysis & Design
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Organization / Project Owner	Jumuia Digital
Planned Duration	Six months
Planning Budget	KSh 4,600,000 (proposed estimate)

Project context

Jumuia X is a service discovery and booking platform intended to help customers find, compare, and book service providers while helping businesses manage branches, staff, services, pricing, appointments, and basic performance information. The platform already has a working implementation; this baseline plan formalizes the work required to refine, test, validate, and prepare a broader production rollout.

Prepared for academic planning and project evaluation

1. Executive Summary

This Baseline Project Plan (BPP) defines the proposed scope, objectives, approach, schedule, resources, budget, feasibility considerations, and risks for the next planned phase of Jumuia X. The project addresses two connected problems: customers can find it difficult to discover and compare suitable service providers and arrange appointments, while service businesses may rely on fragmented or manual processes to coordinate staff, services, prices, and bookings.

The proposed solution is an integrated web-based system that supports customer accounts, business and branch information, service catalogues and pricing, staff coordination, booking workflows, discovery, ratings, and basic management reporting. The system is primarily a Transaction Processing System (TPS), with selected Management Information System (MIS) capabilities through summaries and reports.

The plan uses an iterative and incremental Agile approach supported by standard System Development Life Cycle (SDLC) documentation. The proposed duration is six months and the preliminary planning budget is KSh 4,600,000. These figures are estimates for planning, not evidence that the full amount has been spent or that all costs have been contractually confirmed.

2. Project Background and Problem Statement

Service customers often depend on social media, personal referrals, phone calls, or in-person visits to learn what a provider offers, compare prices, check availability, and make appointments. Information can be incomplete or scattered across different channels. A customer who is unfamiliar with a town may have particular difficulty identifying a suitable and trustworthy provider.

On the business side, bookings, staff schedules, service lists, prices, and daily activity may be handled through separate conversations, notebooks, spreadsheets, or social media accounts. These fragmented processes can cause missed messages, scheduling conflicts, inconsistent information, and limited visibility into business activity. Jumuia X proposes one coordinated digital platform for discovery, booking, and essential business operations.

3. Project Objectives

- Provide customers with a central place to discover service businesses, review service information, and make or manage bookings.
- Enable business owners and authorized staff to maintain branch details, services, prices, and staff information within defined permissions.
- Support a consistent booking lifecycle, including booking creation, status updates, and cancellation where applicable.
- Provide ratings and basic reporting to support customer confidence and business decision-making.
- Improve operational consistency through validation, access controls, testing, documentation, and a controlled production deployment.
- Deliver a maintainable foundation that can be expanded in later phases when demand, cost, and compliance requirements are better validated.

4. Project Scope

In scope

- Customer registration, authentication, profile, and account access controls.
- Business, branch, staff, service, and pricing management.
- Service discovery and display of relevant provider information.
- Booking creation and core booking status/cancellation workflows.
- Ratings, basic management summaries, and operational reporting.
- Security and validation checks, testing, deployment configuration, documentation, and user orientation.

Out of scope for this baseline

- Advanced AI-powered recommendations and predictive analytics.
- A mature, end-to-end platform payment system; payment collection and settlement require separate business, technical, and compliance decisions.
- Immediate rollout of every possible service category or all future marketplace features.
- Nonessential promotional, rewards, or loyalty features unless approved through scope control.

The baseline covers approximately 65% of the broader product vision. This is a planning estimate of the intended product scope, not a measured completion percentage. Any significant change must be assessed for impact on time, cost, risk, and quality before approval.

5. Proposed System and Development Approach

Jumuia X will be treated primarily as a Transaction Processing System because it records and updates operational transactions such as accounts, service selections, appointments, and booking statuses. Its reporting features provide a supporting MIS function by summarizing operational data for business owners and managers.

Development will proceed in short iterations. Each iteration will prioritize a manageable set of requirements, implement the relevant functionality, test it, review the result, and incorporate feedback. SDLC practices will still be used to document requirements, design decisions, tests, deployment, and user guidance. This combines flexibility with the traceability needed for an academic systems-analysis project.

Key functional requirements

- Users can securely register, authenticate, and access functions permitted by their role.
- Authorized business users can manage branch information, service offerings, prices, and staff records. • Customers can discover providers, view service details, and submit booking requests for available services and times. • The system records booking changes and exposes appropriate status information to relevant users. • Customers can submit ratings under the platform's rules, and businesses can access basic operational summaries. **Key non-functional requirements**

- Security: role-based authorization, protected credentials, input validation, secure configuration, and appropriate session/token handling.
- Reliability: predictable booking state changes, validation of conflicting requests, and tested recovery/back-up procedures.
- Performance: efficient database access and responsive core workflows under the expected initial workload. • Usability: mobile-friendly screens, understandable status messages, and role-appropriate navigation. • Maintainability: modular code, clear separation of responsibilities, automated tests, and practical technical documentation.

6. Feasibility Analysis

Technical	Feasible in principle because the system uses established web, database, and cloud technologies. Feasibility depends on integration testing, security review, reliable deployment, and suitable operational monitoring.
Economic	The proposed KSh 4.6 million budget gives a planning baseline. Actual viability requires validating the cost assumptions, expected adoption, operating costs, and a credible route to sustainable revenue or funding.
Operational	The platform can support customer and business workflows if owners, managers, staff, and customers understand the processes. Pilot feedback, onboarding, and training are needed.
Schedule	Six months is a proposed target. It is achievable only with controlled scope, timely decisions, sufficient development and testing capacity, and early resolution of dependencies.
Legal / contractual	Privacy, data protection, terms, communications, vendor terms, business registration, and any applicable regulatory obligations must be reviewed and confirmed before launch. This plan does not claim that registrations or approvals are complete.
Stakeholder / political	Support from service businesses and customers is essential. Stakeholder concerns, trust, data handling, and clarity of responsibilities must be addressed through consultation and transparent communication.

Overall feasibility is conditional rather than automatic: the project should proceed through staged validation, with scope and launch decisions based on evidence from testing, stakeholders, costs, and compliance review.

7. Project Schedule and Milestones

Month 1	Initiation, stakeholder consultation, requirements review, scope baseline, architecture and detailed planning.	Approved scope, requirements and delivery plan.

Months 2–3	Implement and refine core backend services and APIs; data validation, permissions, booking logic, and automated tests.	Core backend workflows tested.
Months 4–5	Frontend refinement and integration; discovery, ratings, management views, reporting, usability checks.	Integrated release candidate.
Month 6	System, security, and user acceptance testing; resolve critical defects; deploy, document, train, and prepare rollout.	Launch readiness review and controlled release.
Post-launch	Monitor performance and incidents, collect feedback, maintain backups, and prioritize approved improvements.	Maintenance and improvement backlog.

8. Preliminary Budget Estimate

The following is a proposed planning estimate in Kenyan shillings. It is not an audited expenditure report, vendor quotation, or confirmation that the services have already been purchased. Actual costs should be validated before commitments are made.

Project management and systems analysis	450,000
Backend engineering and API work	850,000
Frontend development and integration	600,000
UI/UX design	250,000
Cloud infrastructure, database, backups and monitoring	300,000
AI development subscriptions (OpenAI and Claude)	240,000
SMS and OTP communications (Africa's Talking)	180,000
Secure company email and setup	25,000
Business registration, legal and applicable compliance provision	250,000
Security testing, QA and user acceptance testing	300,000
Deployment, domain and production configuration	180,000
Documentation, training and rollout	150,000
Contingency reserve	825,000
Total proposed budget	**4,600,000**

Budget assumptions and validation: cloud, messaging, AI subscriptions, legal/compliance, and rollout costs depend on usage, vendor pricing, and the actual launch footprint. A reference Proton Mail Plus price of US$47.88 per year would be approximately KSh 6,224 at an illustrative exchange rate of KSh 130 per US dollar; this is only a reference for one email plan, not a confirmed total email or communications cost. The KSh 25,000 line includes broader setup assumptions and must be validated. The compliance allocation is a provision, not a statement of any official fee or completed registration.

9. Project Risks and Mitigation

Scope expansion	Schedule and budget overrun.	Use a prioritized backlog, documented change requests, and approval of changes with material impact.
Security or privacy issue	Loss of trust, service interruption, or legal exposure.	Apply least-privilege access, secure configuration, testing, backups, and review of privacy and data-handling obligations.
Cloud/vendor cost growth	Operating costs exceed assumptions.	Set usage limits and alerts, review bills, monitor resource use, and validate alternatives before scaling.

Low user adoption	Limited business value and weak marketplace activity.	Pilot with customers and providers, simplify onboarding, collect feedback, and demonstrate practical value.
Schedule or integration delays	Late release or reduced test time.	Deliver incrementally, test integrations early, track dependencies, and protect time for QA and user acceptance.
Budget uncertainty	Funding gap or unplanned commitments.	Obtain quotes, track actual versus planned costs, use contingency only with approval, and review forecasts regularly.

10. Quality Assurance, Governance and Success Criteria

Quality will be managed through requirements review, automated tests, integration checks, security-focused testing, user acceptance testing, and a controlled release process. Critical defects affecting authentication, access control, booking correctness, or data integrity must be resolved or formally assessed before launch. Deployment should include monitoring, backups, recovery procedures, and a clear process for reporting and responding to incidents.

- Core customer and business workflows pass agreed functional and user acceptance tests.
- Role permissions and validation prevent unauthorized or inconsistent operations in tested scenarios.
- No unresolved critical defects remain at the launch decision; remaining lower-priority defects are documented and accepted by the responsible stakeholders.
- Business users can maintain key information and customers can complete the intended booking journey with understandable feedback.
- Deployment, backup, support, and user documentation are available to the people responsible for

operation.

11. Conclusion

Jumuia X is intended to make service discovery and booking more coordinated for customers while giving service businesses a more consistent way to manage essential operations. This BPP establishes a proposed six-month delivery path, a preliminary KSh 4,600,000 budget, a controlled scope, and practical feasibility and risk checks. Progress should be reviewed at each milestone, and rollout should depend on tested functionality, stakeholder feedback, validated costs, and completion of applicable legal and privacy reviews.

Project repository

The academic assignment repository is available at github.com/KiplagatKeith/System-Analysis-Design-Mid-sem-Assignment.